

# Supplementary information

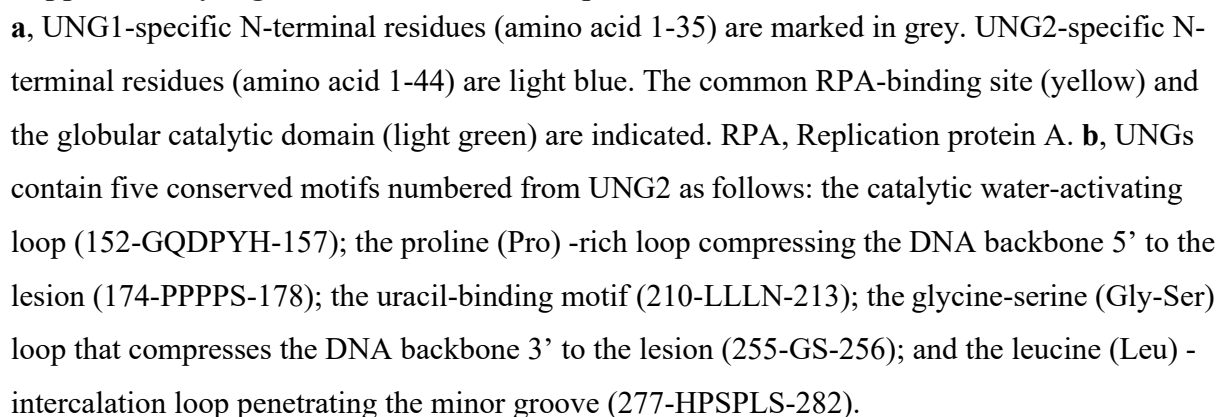
## Supplementary Figures

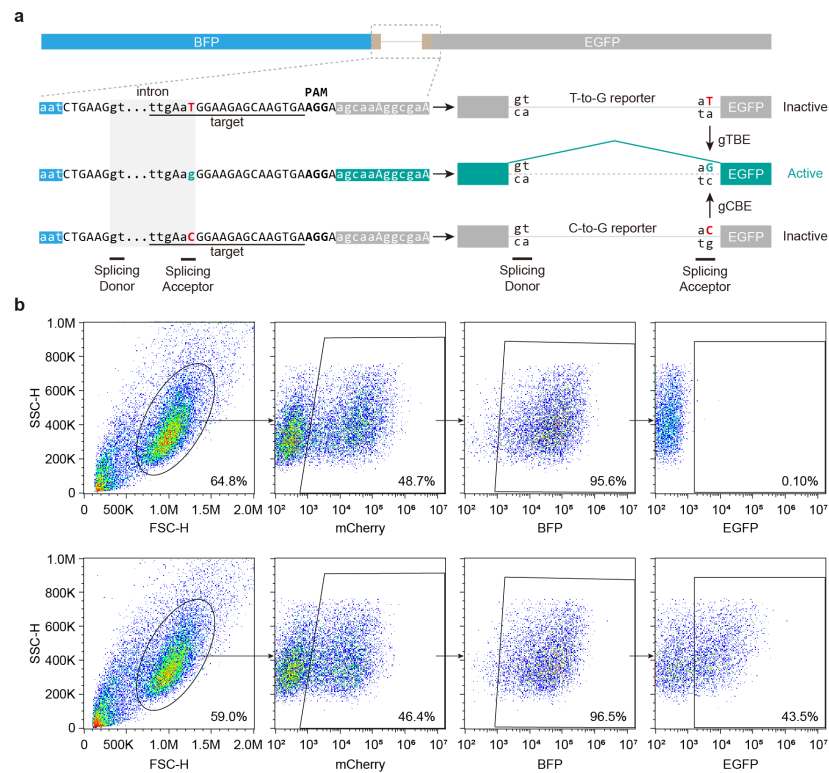
- Supplementary Figure 1. Characteristic sequences and motifs of human UNG1 and UNG2.
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## Supplementary Tables

- Supplementary Table 1. Protein sequence of gTBEv3 and gTBEv5 editor.

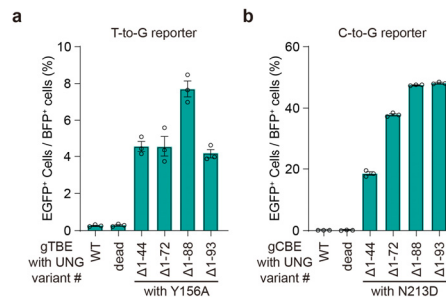
**Supplementary Figure 1. Characteristic sequences and motifs of human UNG1 and UNG2.**





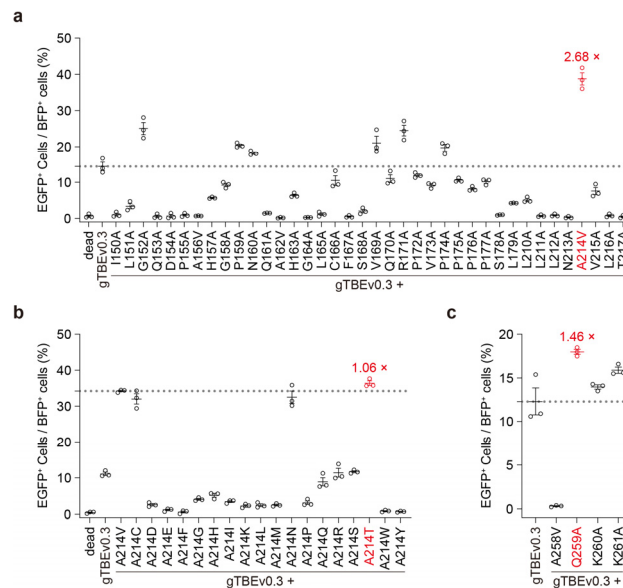
**Supplementary Figure 2. Characterizations of T-to-G and C-to-G reporter system.**

**a**, Schematic construct designs of the reporter for T-to-G or C-to-G editing detection. PAM, Protospacer adjacent motif. **b**, Representative flow cytometry scatter plots showing gating strategy and the percentages of gated cells for the negative control (upper panel) and gCBEv0.3 (lower panel).



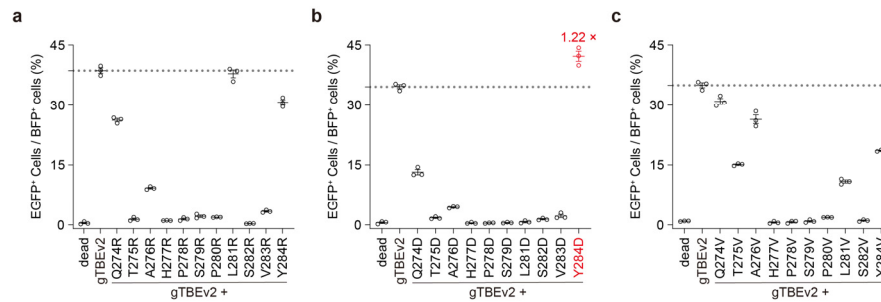
**Supplementary Figure 3. Editing activity of gTBE and gCBE candidates with various UNG-NTD truncations.**

**a**, Percentage of EGFP<sup>+</sup> cells for evaluating T editing activity of gTBE candidates with various UNG variants ( $n = 3$  independent biological replicates). **b**, Percentage of EGFP<sup>+</sup> cells for evaluating C editing activity of gCBE candidates with various UNG variants ( $n = 3$  independent biological replicates). WT, wild-type UNG2 $\Delta$ 88. dead, catalytically inactive UNG2 $\Delta$ 88 (carrying D154N and H277N mutations, equivalent to D145N and H268N of UNG1). All values are presented as mean  $\pm$  s.e.m. Source data are provided as a Source Data file.

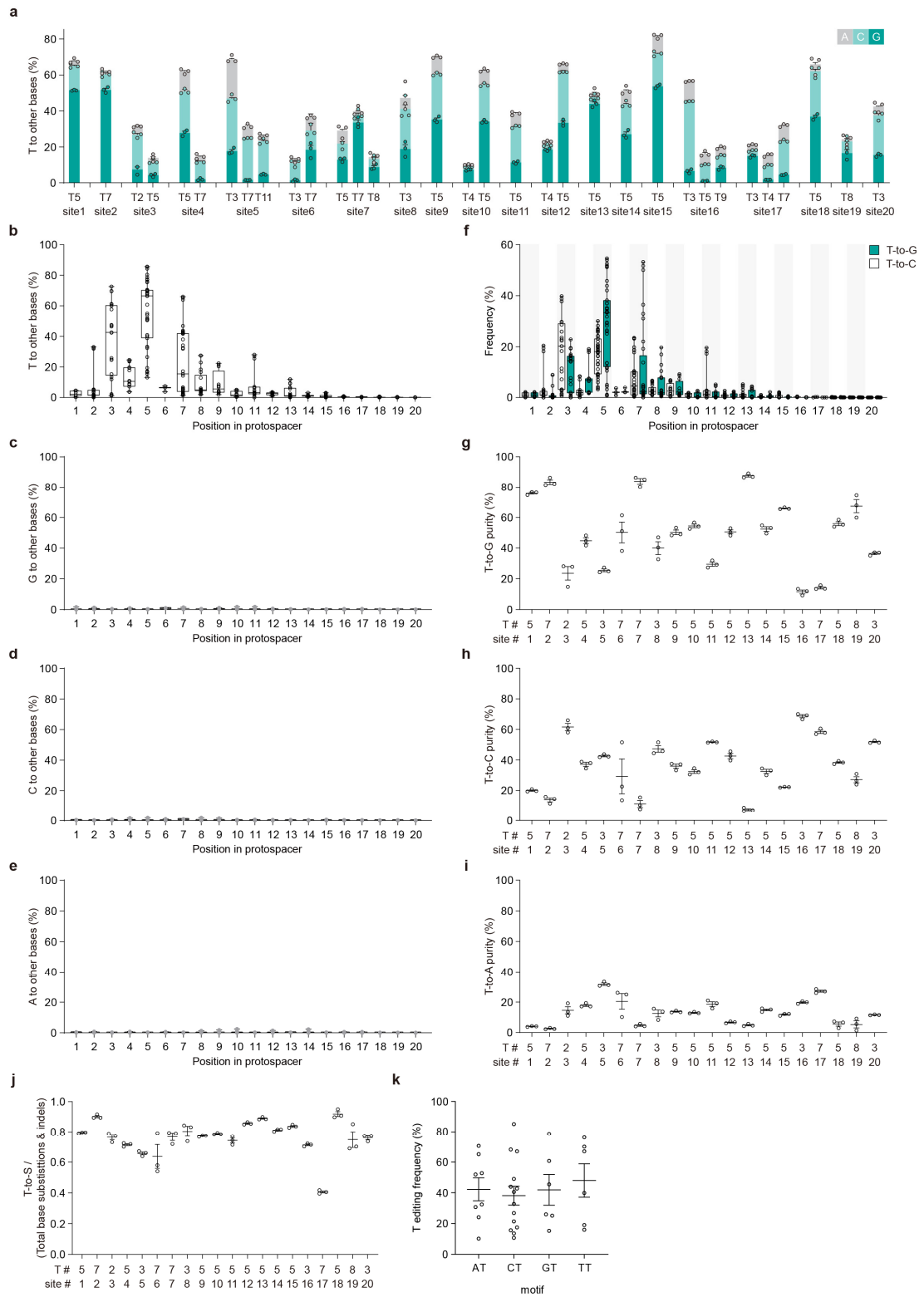


**Supplementary Figure 4. Performance of engineered variants in the gTBEv0.3 background.**

**a**, Percentage of EGFP<sup>+</sup> cells of gTBE variants from alanine-scanning mutagenesis of regions covering the catalytic water-activating loop, the Pro-rich loop, and the uracil-binding motif ( $n = 3$  independent biological replicates). Replacement of alanine with valine, A>V, is intended to cover all the residues in the interested regions. **b**, Percentage of EGFP<sup>+</sup> cells of gTBE variants from site-saturation mutagenesis of the residue at position 214 ( $n = 3$  independent biological replicates). **c**, Percentage of EGFP<sup>+</sup> cells of gTBE variants with mutations of selected spatial neighbors of residue T214 ( $n = 3$  independent biological replicates). dead, catalytically inactive UNG2Δ88 (carrying D154N and H277N mutations, equivalent to D145N and H268N of UNG1). All values are presented as mean ± s.e.m. Source data are provided as a Source Data file.



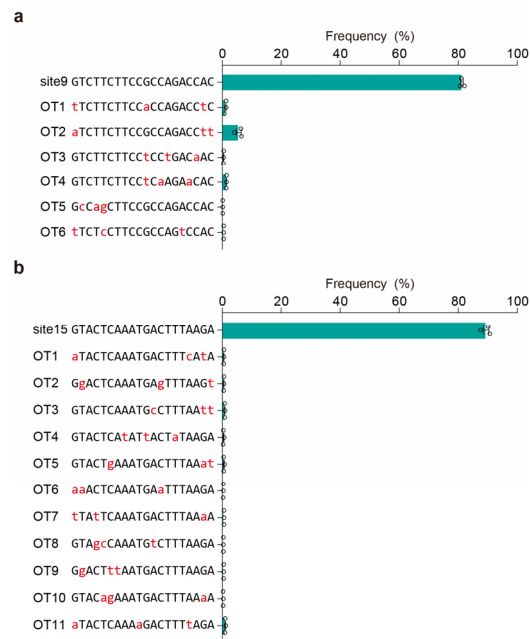
**Supplementary Figure 5. Performance of engineered variants in the gTBEv2 background.** **a-c**, Percentage of EGFP<sup>+</sup> cells of gTBE candidates with different UNG variants from sequential substitutions of arginine (**a**), aspartic acid (**b**), and valine (**c**) (X>R, D, or V) ( $n = 3$  independent biological replicates). dead, catalytically inactive UNG2 $\Delta$ 88 (carrying D154N and H277N mutations, equivalent to D145N and H268N of UNG1). All values are presented as mean  $\pm$  s.e.m. Source data are provided as a Source Data file.



**Supplementary Figure 6. Further characterization of editing profiles for gTBEv3.**

**a**, Stacked bar plots showing the on-target DNA base editing at positions with T conversion frequencies >10% at each genomic site in HEK293T cells (mean  $\pm$  s.e.m.,  $n = 3$  independent biological replicates). **b-e**, Frequencies of T (**b**), G (**c**), C (**d**) and A (**e**) conversions by gTBEv3 across the protospacer positions 1-20 (where PAM is at positions 21–23) from the edited sites in Figure 3a. **f**, Frequencies of T-to-G and T-to-C editing by gTBEv3. In **b-f**, single dot represents individual replicate ( $n = 3$  independent replicates per site), and boxes span the interquartile range (25th to 75th percentile); horizontal lines within the boxes indicate the median (50%); and whiskers extend to the minimal and maximal values. **g-i**, Percentage of T-to-G (**g**), T-to-C (**h**) or T-to-A (**i**) editing by gTBEv3 at various edited sites shown in Figure 3a (mean  $\pm$  s.e.m.,  $n = 3$  independent biological replicates). T#: T position with highest on-target base editing frequencies across protospacer positions 1–20. site #: genomic site number. **j**, The ratio of T-to-S to total T editing (base conversions and indels) by gTBEv3 editing at the sites shown in Figure 3a. **k**, The statistical analysis of on-target DNA base editing for each NT motif from the edited sites in (**a**). Each dot represents the mean of three biological replicates for each edited position at various edited sites.  $n = 8, 14, 6, 6$  for motif AT, CT, GT, TT. Source data are provided as a Source Data file.

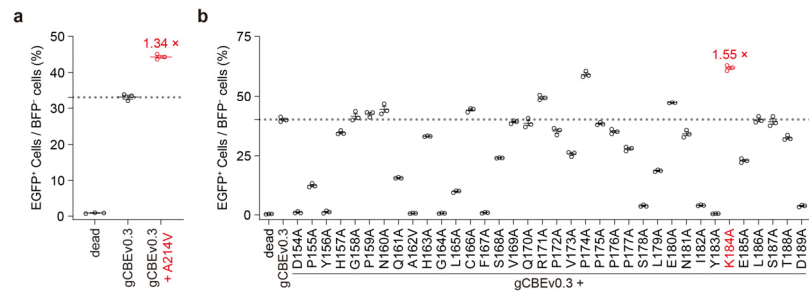




### Supplementary Figure 7. The sgRNA-dependent off-target analysis for gTBEv3 at more sites.

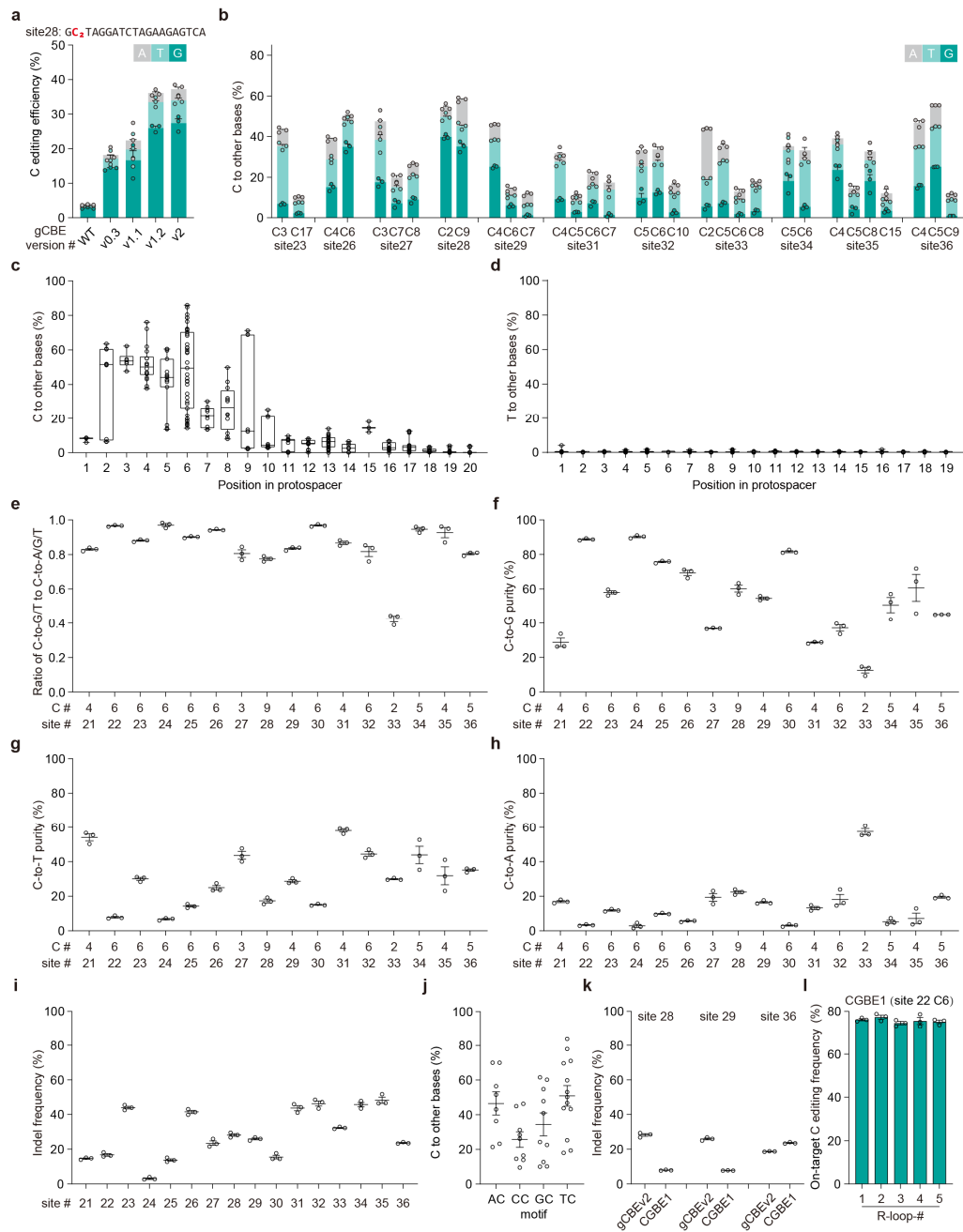
The sgRNA-dependent off-target analysis for gTBEv3 editing at site 9 (**a**) and site 15 (**b**) ( $n = 3$  independent biological replicates). OT: off-target. All values are presented as mean  $\pm$  s.e.m.

Source data are provided as a Source Data file.



### Supplementary Figure 8. Performance of engineered variants in the gCBEv0.3 background.

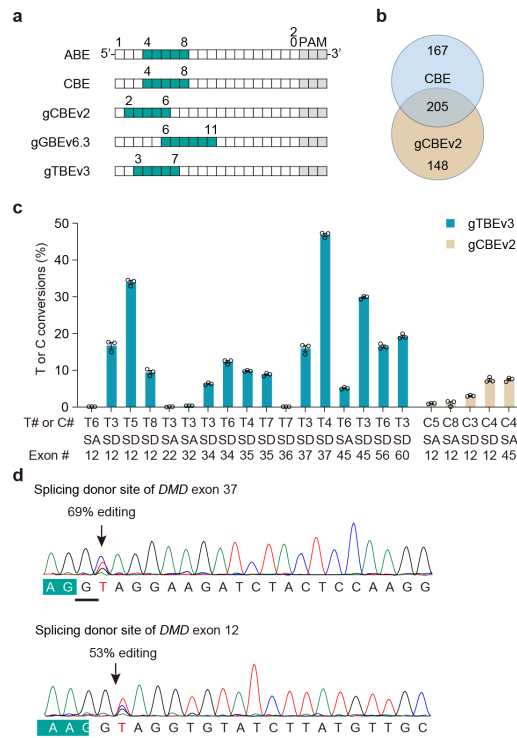
**a**, Percentage of EGFP<sup>+</sup> cells of gCBE variant by introduction of the mutation A214V ( $n = 3$  independent biological replicates). **b**, Percentage of EGFP<sup>+</sup> cells of gCBE variants from alanine-scanning mutagenesis of regions covering the catalytic water-activating loop and the Pro-rich loop ( $n = 3$  independent biological replicates). Replacement of alanine with valine (A>V) is intended to cover all the residues in the interested regions. dead, catalytically inactive UNG2Δ88 (carrying D154N and H277N mutations, equivalent to D145N and H268N of UNG1). All values are presented as mean  $\pm$  s.e.m. Source data are provided as a Source Data file.



### Supplementary Figure 9. Further characterization of editing profiles for gCBEv2.

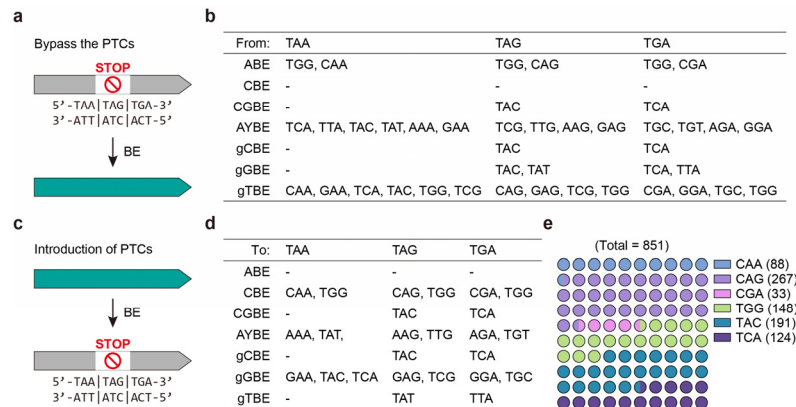
**a**, Frequency of C base editing outcomes with different gCBE variants at the edited C2 position in site 28 in transfected HEK293T cells by target deep sequencing (mean  $\pm$  s.e.m.,  $n = 3$  independent biological replicates). **b**, Stacked bar plots showing the on-target DNA base editing at two or more positions with C conversion frequencies  $>10\%$  at each genomic site in HEK293T cells (mean  $\pm$  s.e.m.,  $n = 3$  independent biological replicates). **c,d**, Frequencies of C (**c**) and T (**d**)

conversions by gCBEv2 across the protospacer positions 1-20 (where PAM is at positions 21–23) from the edited sites in Figure 4c. Single dot represents individual replicate ( $n = 3$  independent replicates per site), and boxes span the interquartile range (25th to 75th percentile); horizontal lines within the boxes indicate the median (50%); and whiskers extend to the minimal and maximal values. **e**, The ratio of C-to-G/T to C-to-A/G/T conversion frequency by gCBEv2 editing at the sites shown in Figure 4c. **f-h**, Percentage of C-to-G (**f**), C-to-T (**g**) or C-to-A (**h**) editing by gCBEv2 at various edited sites shown in Figure 3a (mean  $\pm$  s.e.m.,  $n = 3$  independent biological replicates). **i**, indels frequencies with gCBEv2 at 16 on-target sites (mean  $\pm$  s.e.m.,  $n = 3$  independent biological replicates). In **e-i**, C#: C position with highest on-target base editing frequencies across protospacer positions 1-20. site #: genomic site number. **j**, The statistical analysis of on-target DNA base editing for each NC motif from the 16 edited sites. Each dot represents the mean of three biological replicates for each edited position at various edited sites.  $n = 8, 9, 10, 13$  for motif AC, CC, GC, TC. **k**, indels frequencies with gCBEv2 and CGBE1 at 3 on-target sites from Figure 4d (mean  $\pm$  s.e.m.,  $n = 3$  independent biological replicates). **l**, On-target base editing frequencies for CGBE1 at C6 of site 22 in HEK293T cells for the orthogonal R-loop assay (mean  $\pm$  s.e.m.,  $n = 3$  independent biological replicates). Source data are provided as a Source Data file.



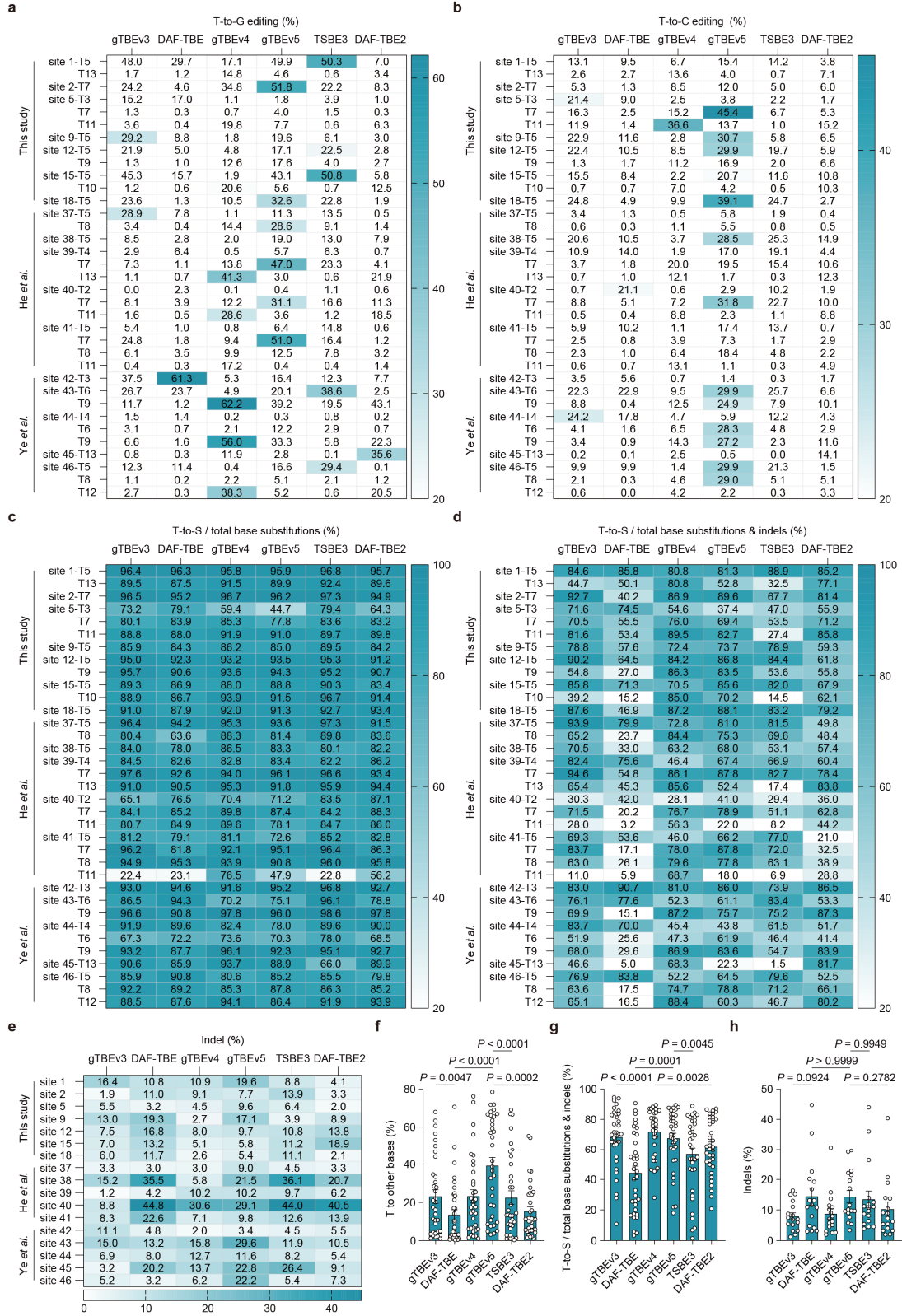
### Supplementary Figure 10. Base editing at splicing sites with gTBEv3.

**a**, The optimal editing windows for various base editors. PAM, Protospacer adjacent motif. **b**, Venn diagram showing the distribution of sgRNAs for CBE and gCBEv2 in Figure 5b. **c**, Bar plots showing the frequency of T or C conversions at several splicing acceptor (SA) or splicing donor (SD) sites of interest targeted by gTBEv3 or gCBEv2 (mean  $\pm$  s.e.m.,  $n = 3$  independent biological replicates). T# or C#: The position of targeted T or C across protospacer positions 1–20. **d**, DNA sequencing chromatograms for targeting the SD site of human *DMD* exon 37 and exon 12 with gTBEv3. Sanger sequencing results were quantified by EditR. Source data are provided as a Source Data file.



### Supplementary Figure 11. PTCs editing and introduction for various base editors.

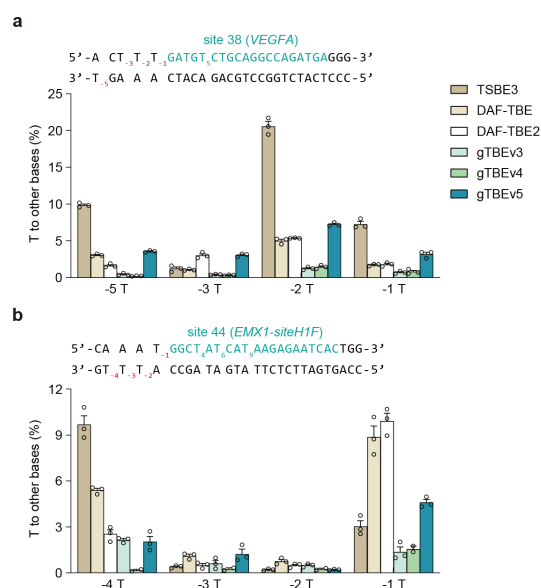
**a**, principle for bypassing premature termination codons (PTCs) with various base editors. **b**, the possible codon outcomes from stop codons (TAA, TAG or TGA) editing with different base editors. **c**, principle for introduction of PTCs with various base editors. **d**, the available codons for editing into stop codons (TAA, TAG or TGA) with different base editors. **e**, The  $10 \times 10$  dot plot diagram showing the percentage of possible sgRNAs for introduction of premature termination codons (PTCs) by targeting different codons (with the number of available sgRNAs presented in the right) in 15 well-studied genes (*AGT*, *ANGPTL3*, *APOC3*, *B2M*, *CD33*, *DNMT3A*, *HPD*, *KLKB1*, *PCSK9*, *PDCD1*, *PRDMI*, *TGFBR2*, *TRAC*, *TTR*, *VEGFA*) for gene and cell therapy research with gGBEv6.3 and CBE. In **b** and **d**, AYBE, AYBEv3; gCBE, gCBEv2; gGBE, gGBEv6.3; gTBE, gTBEv3. Source data are provided as a Source Data file.



**Supplementary Figure 12. Additional comparison of different gTBEs.**

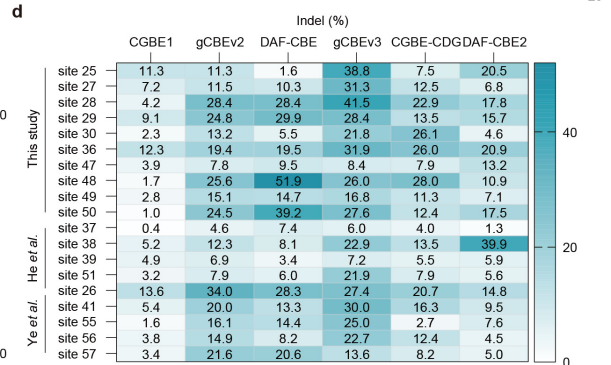
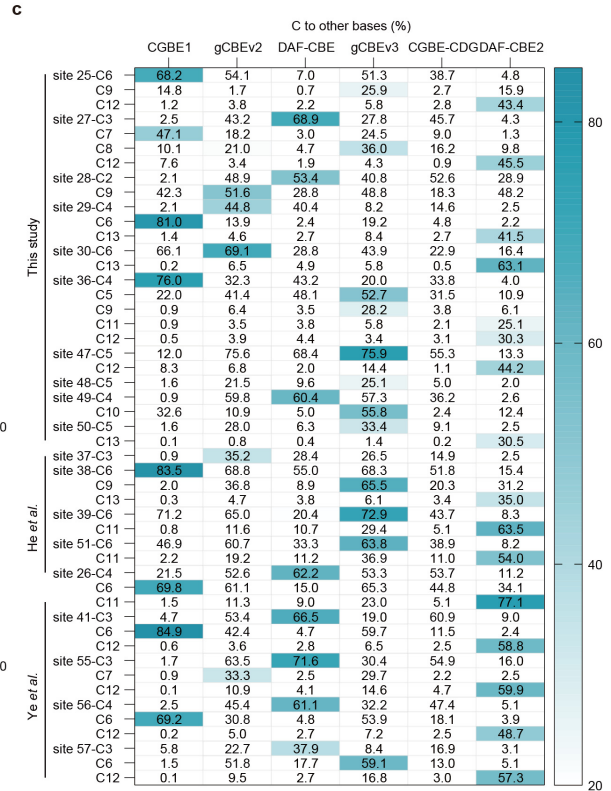
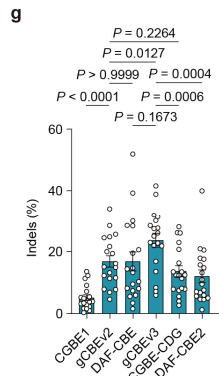
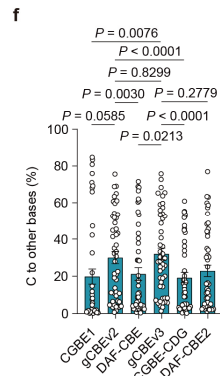
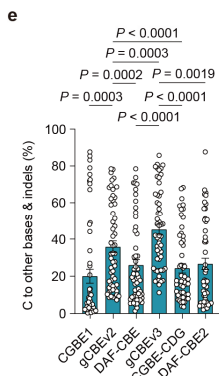
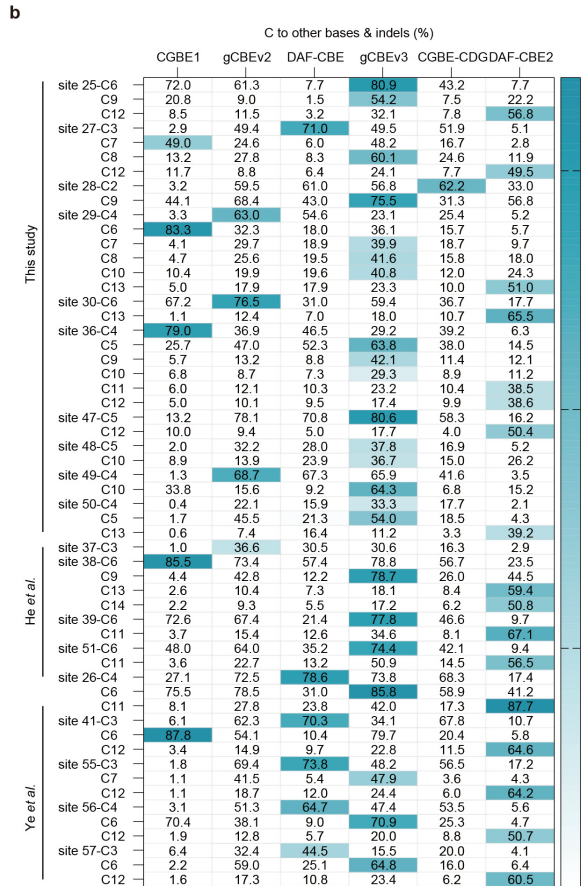
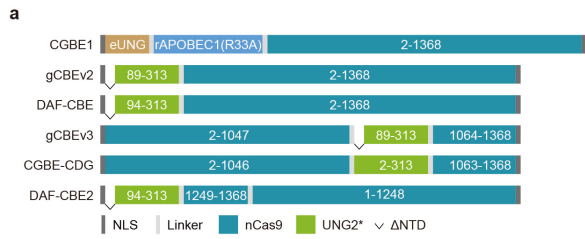
**a,b**, The frequencies of T-to-G (**a**) or T-to-G (**b**) conversions. The highest frequencies (>20%) of edited thymines at corresponding positions were highlighted as Heat map. **c,d**, Heat map showing the percentages of T-to-S to total base conversions (**c**) or T-to-S to total T editing (base conversions and indels; **d**) for various base editors. **e**, Heat map showing the indels frequencies for various base editors. For **a-e**,  $n = 3$  independent biological replicates per site, note  $n = 2$  for site 44 targeted by gTBEv4. **f-h**, The statistical analysis of T base editing (**f**,  $n = 35$  positions), T-to-S percentages (**g**,  $n = 35$  positions) and indels (**h**,  $n = 17$  sites). All values are presented as mean  $\pm$  s.e.m. Each dot represents the mean of three biological replicates for each edited position at various edited sites. **a-h**, the graphs were derived from the data for various base editors shown in Figure 6c. Dunnett's multiple comparisons test after one-way ANOVA was used to compare the gTBEv3 or gTBEv5 with other base editors in f-h. Source data are provided as a Source Data file.





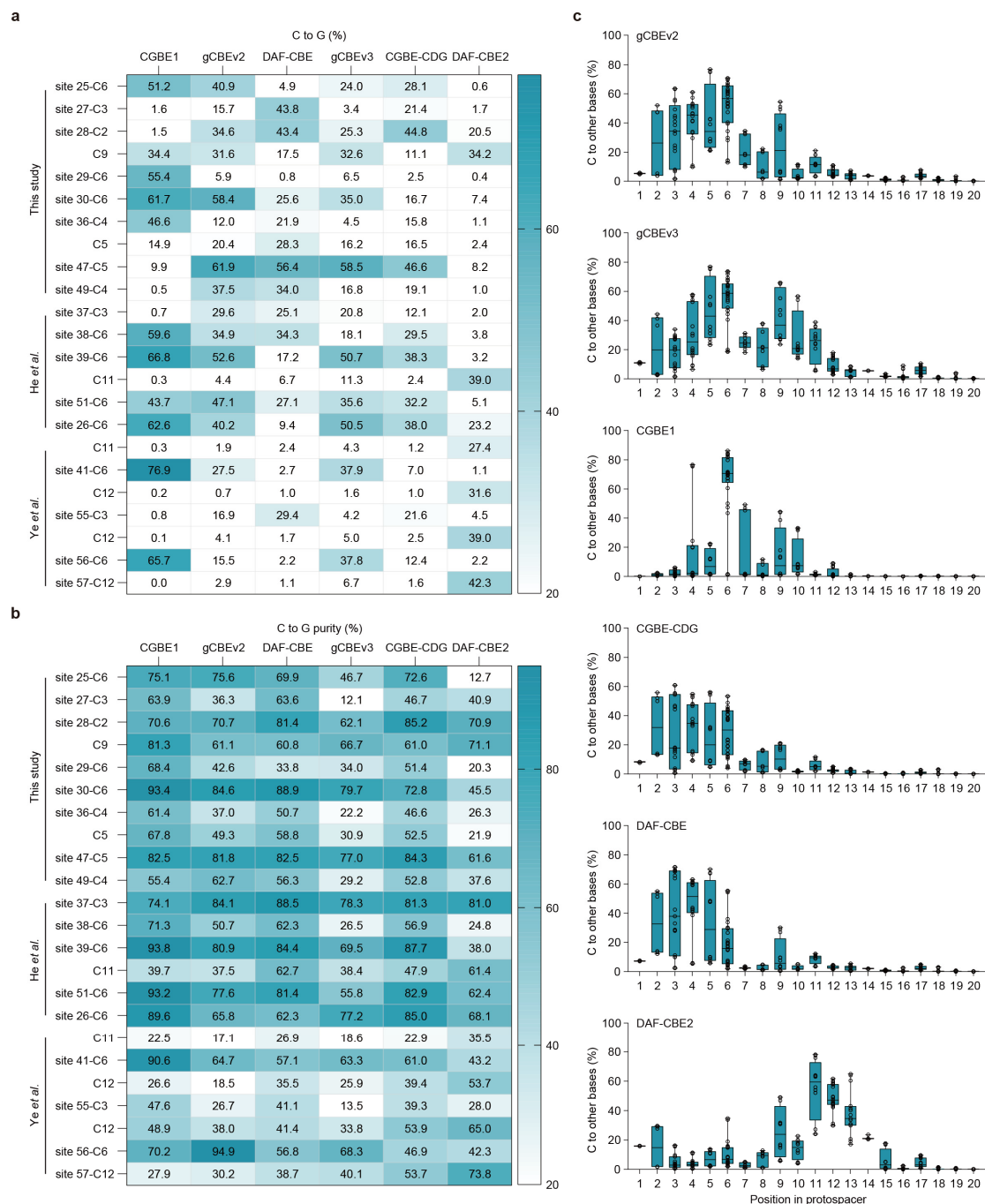
**Supplementary Figure 13. T editing in the dsDNA upstream from the target site.**

**a**, Bar plots showing the frequency of T conversation at -5T, -3T, -2T and -1T in the dsDNA upstream from the target site 38 (*VEGFA*) ( $n = 3$  independent biological replicates). **b**, Bar plots showing the frequency of T conversation at -4T, -3T, -2T and -1T in the dsDNA upstream from the target site 44 (*EMX1-siteH1F*) ( $n = 3$  independent biological replicates, note  $n = 2$  for gTBEv4). All values are presented as mean  $\pm$  s.e.m., the graphs were derived from the data for various base editors shown in Figure 6c. Source data are provided as a Source Data file.



**Supplementary Figure 14. Comparison of various glycosylase-based base editors for cytosine editing.**

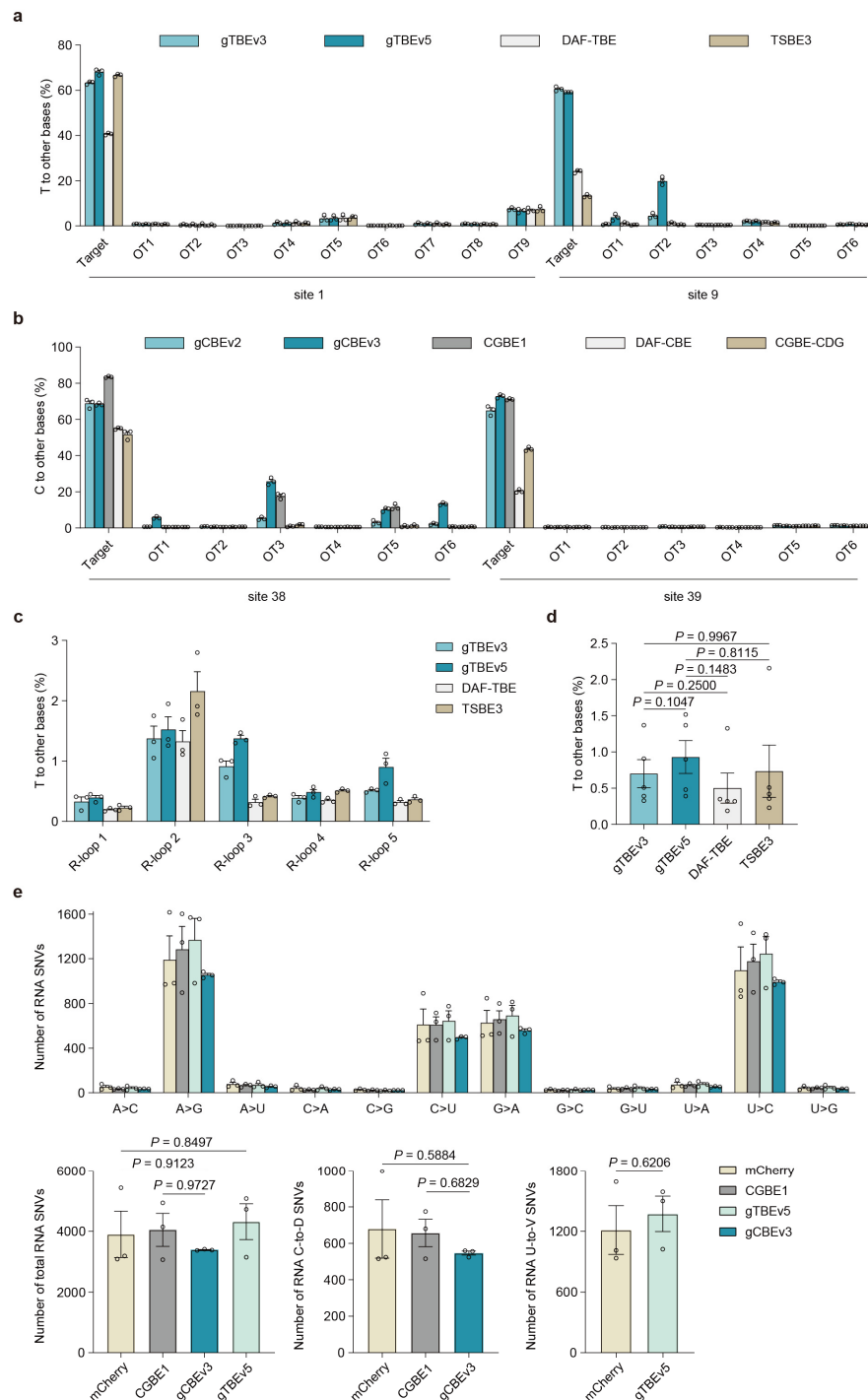
**a**, Schematic of the basic architectures for various base editors. The bipartite nuclear localization signal (bpNLS) is shown in dark gray, linker in light gray, and nCas9 in teal green. UNG2\* (in light green), UNG2 variant from the corresponding base editor.  $\Delta$ NTD, deletion of the N-terminal domain. **b,c**, The frequencies of total C editing (base conversions and indels, **b**) or C base conversions (**c**) for various base editors at 19 endogenous loci. The cytosines with editing frequencies >25% for any base editors were showed. The highest frequencies at corresponding positions were highlighted as Heat map ( $n = 3$  independent biological replicates). **d**, Heat map showing the indels frequencies for various base editors ( $n = 3$  independent biological replicates). **e-g**, The statistical analysis of total C editing (**e**,  $n = 56$  positions), C base conversions (**f**,  $n = 49$  positions) and indels (**g**,  $n = 19$  sites). All values are presented as mean  $\pm$  s.e.m. Each dot represents the mean of three biological replicates for each edited position at various edited sites. Dunnett's multiple comparisons test after one-way ANOVA was used to compare the gCBEv2 or gCBEv3 with other base editors. Source data are provided as a Source Data file.



**Supplementary Figure 15. Additional comparison of various glycosylase-based base editors for cytosine editing.**

**a,b**, Heat map showing the C-to-G frequencies (**a**) or C-to-G purity (**b**) for various base editors ( $n = 3$  independent biological replicates). The cytosines with C-to-G editing frequencies  $>25\%$  for any base editors were highlighted. **c**, Frequencies of C base conversions by various base

editors across the protospacer positions 1-20 (where PAM is at positions 21–23). Single dot represents individual replicate ( $n = 3$  independent replicates per site), and boxes span the interquartile range (25th to 75th percentile); horizontal lines within the boxes indicate the median (50%); and whiskers extend to the minimal and maximal values. The graphs were derived from the data for various base editors shown in Supplementary Figure 14c. Source data are provided as a Source Data file.



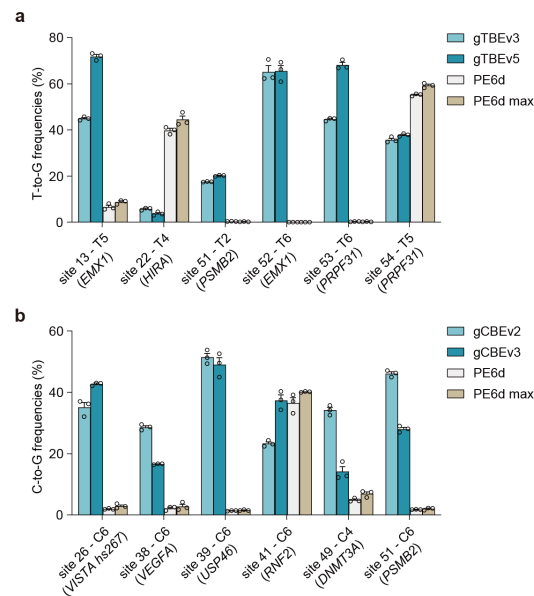
**Supplementary Figure 16. Off-target analysis of various glycosylase-based base editors.**

**a,b**, The sgRNA-dependent off-target analysis of cumulative T editing (**a**) or C editing (**b**)

frequencies for various base editors at corresponding sites ( $n = 3$  independent biological

replicates). OT: off-target. **c**, The sgRNA-independent cumulative off-target T editing detected

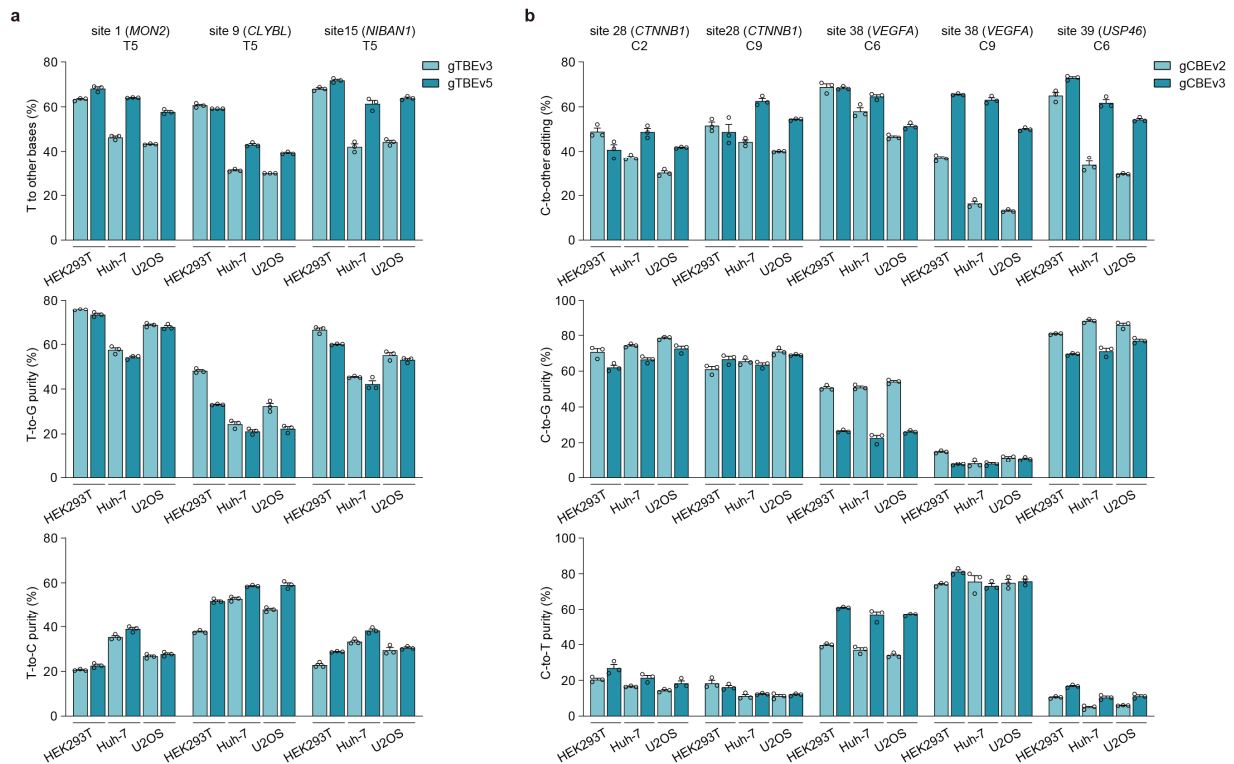
by the orthogonal R-loop assay at each R-loop site ( $n = 3$  independent biological replicates). **d**, The statistical analysis of sgRNA-independent off-target T editing ( $n = 5$  sites). Each dot represents the mean of three biological replicates for each edited position at various edited sites. Dunnett's multiple comparisons test after one-way ANOVA was used to compare the gTBEv3 or gTBEv5 with other base editors. **e**, RNA off-target analysis for various base editors ( $n = 3$  independent biological replicates). The mCherry was used as control. D = A or G or U; V = A or C or G. For multiple comparisons, the Dunnett's multiple comparisons test after one-way ANOVA was used to compare the gCBEv3 or gTBEv5 with other groups. For comparison of RNA U-to-V SNVs for gTBEv5 and mCherry, two-tailed unpaired two-sample  $t$  test was used. All values are presented as mean  $\pm$  s.e.m. Source data are provided as a Source Data file.



### Supplementary Figure 17. Comparison between gTBEs or gCBEs and PEs.

**a**, Bar plots showing the on-target T-to-G editing frequency for various editors at each genomic site in HEK293T cells ( $n = 3$  independent biological replicates). OT: off-target. **b**, Bar plots showing the on-target C-to-G editing frequency for various editors at each genomic site in HEK293T cells ( $n = 3$  independent biological replicates). The PE6d was used together with epegRNA and nick sgRNA. For PE6d max, PE6d was co-expressed with the codon-optimized hMLH1dn, a dominant negative MMR protein. All values are presented as mean  $\pm$  s.e.m. Source data are provided as a Source Data file.





**Supplementary Figure 18. Characterization of editing profiles for gTBEs or gCBEs in HEK293T, Huh-7, and U2OS cells.**

**a**, Bar plots showing the on-target T base editing frequency, T-to-G editing purity or T-to-C editing purity for gTBEv3 and gTBEv5 in different cell lines ( $n = 3$  independent biological replicates). **b**, Bar plots showing the on-target C base editing frequency, C-to-G editing purity or C-to-T editing purity for gCBEv2 and gCBEv3 in different cell lines ( $n = 3$  independent biological replicates). Huh-7, a cell line established from a human hepatocellular carcinoma; U2OS, a cell line established from a human bone osteosarcoma. All values are presented as mean  $\pm$  s.e.m. Source data are provided as a Source Data file.

## Supplementary Tables

### Supplementary Table 1. Protein sequence of gTBEv3 and gTBEv5 editor.

gTBEv3:

BpNLS-**UN**G2(**Y156A,A214T,Q259A,Y284D**)-Linker-**n**Cas9(**D10A**)-BpNLS  
MKRTADGSEFESPKKKRKVN**NPV**VGFGESWKKHLSGEFGKPYFIKLMGFV  
AEERKH**Y**TVYPPPHQVFTWTQMCDIKDVKV**V**ILGQDP**A**HGPNQA**H**GLC  
FSVQRPVPPPPSLENIYKELSTDIEDFVHPGHGDL**S**GWAKQGVLLLN**T**VLT  
VRAHQANSHKERGWEQFTDAVVSWLNQNSNGLV**F**LLWGSYA**A**KKGSA  
IDRK**R**HHVLQTAHPSPLSV**D**RGGFFGCRHFSKTNELLQKSGKKPIDWKELS  
GGSSGGSSGSETPGTSESATPESSGGSSGGSDKKYSIGLAIGTNSVGWAVI  
TDEYKVPSKKFKVLGNTDRHSIKKNLIGALLFDSGETAEATRLKRTARRR  
YTRRKNRICYLQEIFS**N**EMAKVDD**S**FFHRLEESFLVEEDKKHERHPIFGNI  
VDEVAYHEKYPTIYHLRKKLV**D**STD**K**ADLR**L**IYLA**A**HMIKFRGHFLIEG  
DLNP**D**NSDV**D**KLFIQLVQTY**N**QLFEENPINASGVDAKAILSARLSK**S**RRLE  
NLIAQLPGEKKNGLFGNLI**A**LSLGLTPNFKSNFDLAEDAKLQLSKDTYDD  
DLNLLAQIGDQYADLFLAAKNLS**D**AILLSDILRVNTEITKAPLSASMIKR  
YDEHHQDLTLLKALVRQQLPEKYKEIFFDQSKNGYAGYIDGGASQEEFY  
KFIKPILEKMDGTEELLVKLNREDLLRKQRTFDNGSIPHQIHLGELHAILR  
RQEDFY**P**FLKDNREKIEKILTFRIPYYVGPLARGNSRF**A**WMTRKSEETITP  
WNFEEVVDKGAS**A**QSFIERMTNFDKNLPNEKVLPKHSLLYEYFTVYNEL  
TKVKYVTEGMRKPAFLSGEQKKAIVD**L**LFKTNRKVTVKQLKEDYFKKIE  
CFDSVEISGVEDRFNASLGT**Y**HDLLK**I**IKDKD**F**LDNEENEDILEDIVLTLTL  
FEDREMIEERLKTYAHLFDDKVMKQLKRRRYTG**W**GRLSRK**L**INGIRD**K**Q  
SGKTILDFLKSDGFANR**N**F**M**QLIHDDSLTFKEDIQKAQVSGQGD**S**LHEHI  
ANLAGSPA**I**KKGILQTVKV**V**DELVKVMGRHKPENIV**I**EMARE**N**QTTQKG  
QKNSRERMKRIE**E**G**I**KELGSQILKEHPVENTQLQNEKLYLYYLQNGRDM  
YVDQELDINRLSDYDV**D**HIVPQSFLKDDSIDNKVLTRSDKNRGKSDNVPS  
EEVVKKMKNYWRQL**N**AKLITQRKFDNLTKAERGGLSELDKAGFIK**R**Q  
LVETRQITKHVAQILDSRMNTKYDENDKLIREVKVITLKS**K**LVSDFRKDF  
QFYKVREINNYHHAHDAYLNAVVG**T**ALIKKYPKLESEFVYGDYKVYDV  
RK**M**IAKSEQEIGKATAKYFFYSNIMNFFKTEITLANGEIRKRPLIETNGETG  
EIVWDKGRDFATVRKVLSMPQVNIVKKTEVQTGGFSKESILPKRNSDKLI  
ARKKDWD**P**KKYGGFDSP**T**VAYSVLV**V**AKVEKGKSKKLKSVKELLGITI  
MERSSFEKNPIDFLEAKGYKEVKKDL**I**IKLPKYS**L**FELENGRKRMLASAG  
ELQKG**N**ELALPSKYVNFLYLASHYEKLKGSPEDNEQKQLFVEQH**K**HYLD  
E**I**IEQISEFSKR**V**ILADANLDKVLSAYNKHRDKPIREQAEN**I**IHLFTLTNLG  
APAAFKYFDTTIDRKRYTSTKEVLDATLIHQ**S**ITGLYETRIDLSQLGGD**S**G  
GSKRTADGSEFEPKKRK**V**

gTBEv5:

BpNLS-Linker-nCas9(D10A, 2-1047)-Linker-**UNG2**(Y156A, A214T, Q259A, Y284D)-Linker-nCas9(D10A, 1064-1368)-BpNLS

MKRTADGSEFESPKKKRKVSGGSDKKYSIGLAIGTNSVGWAVITDEYKV  
PSKKFKVLGNTDRHSIKKNLIGALLFDSGETAEATRLKRTARRRYTRRKN  
RICYLQEIFSNEMAKVDDSFHRLSEESFLVEEDKKHERHPIFGNIVDEVAY  
HEKYPTIYHLRKKLVDSTDKADRLIYLALAHMIKFRGHFLIEGDLNPDN  
SDVDKLFQILVQTYNQLFEENPINASGVDAKAILSARLSKSRRLLENLIAQL  
PGEKKNGLFGNLIALSLGLTPNFKSNFDLAEDAKLQLSKDTYDDDLNLL  
AQIGDQYADFLAAKNLSDAILLSDILRVNTEITKAPLSASMIKRYDEHHQ  
DLTLLKALVRQQLPEKYKEIFFDQSKNGYAGYIDGGASQEEFYKFIKPILE  
KMDGTEELLVKLNREDLLRKQRTFDNGSIPHQIHLGELHAILRRQEDFYP  
FLKDNREKIEKILTRIPYYVGPLARGNSRFAWMTRKSEETITPWNFEVV  
DKGASAQSFIERMTNFDKNLPNEKVLPHSLLYEYFTVYNELTKVKYVT  
EGMRKPAFLSGEQKKAIVDLLFKTNRKVTVKQLKEDYFKKIECFDSVEIS  
GVEDRFNASLGTYHDLKKIKDKDFLDNEENEDILEDIVLTLTLFEDREMI  
EERLKTYAHLFDDKVMKQLKRRRYTGWGRLSRKLINGIRDKQSGKTILD  
FLKSDGFANRNFMQLIHDDSLTFKEDIQKAQVSGQGDSLHEHIANLAGSP  
AIKKGILQTVKVVDLVKVMGRHKPENIVIAMARENQTTQKGQKNSRER  
MKRIEIEGKELGSQILKEHPVENTQLQNEKLYLYYLQNGRDMYVDQELDI  
NRLSDYDVDHIVPQSFLKDDSIDNKVLTRSDKNRGKSDNVPSEEVVKKM  
KNYWRQLLNAKLITQRKFDNLTKAERGGLSELDKAGFIKRQLVETRQIT  
KHVAQILDSRMNTKYDENDKLIREVKVITLKSCLVSDFRKDFQFYKPREI  
NNYHHAHDAYLNAVVGTAIIKKYPKLESEFVYGDYKVYDVRKMIKSE  
QEIGKATAKYFFYSNIMNFFKSGGSSGGSSGSETPGTSESATPESSGGSSG  
GSNVPVGFGEWKHLSGEFGKPYFIKLMGFVAEERKHHTVYPPPHQVF  
TWTQMCDIKDVKVVLGQDP**A**HGPNQAHGLCFSVQRPVPPPPSLENIYKE  
LSTDIEDFVHPGHGDLGWAKQGVLLLN**T**VLTVRAHQANSHKERGWEQ  
FTDAVVSWLNQNSNGLVFLWGSY**A**KKGSAIDRKRRHVLQTAHPSPLS  
**V****D**RGFFGCRHFSKTNELLQKSGKKPIDWKELSGGSSGGSSGSETPGTSES  
ATPESSGGSSGGSETNGETGEIVWDKGRDFATVRKVLSPQVNIVKKTE  
VQTGGFSKESILPKRNSDKLIARKKDWDPKKYGGFDSPTVAYSVLVAK  
VEKGKSKKLKSVKELGITIMERSSEFKNPIDFLEAKGYKEVKKDLIKLP  
KYSLFELENGRKRMLASAGELQKGNELALPSKYVNFLYLASHYEKLKGS  
PEDNEQKQLFVEQHKHYLDEIIEQISEFSKRVLADANLDKVLSAYNKHR  
DKPIREQAENIIHLFTLTNLGAPAAFKYFDTTIDRKRYTSTKEVLDTLIHQ  
SITGLYETRIDLSQLGGDSGGGSKRTADGSEFEPKKKKRKV